FITBUDDY

AI Fitness Plan Generator Using Gemini Model

Project Report

FitBuddy is an AI-powered application that generates personalized fitness plans using user information such as fitness goal, experience level, available equipment, workout days, and session duration.

Prepared for Academic / Mini Project Submission

1. Abstract

FitBuddy is an AI-based fitness plan generator designed to help users create structured and personalized exercise plans. The application collects basic user preferences and fitness information and sends the data to a Gemini model with a carefully designed prompt. Gemini generates a practical plan containing exercises, sets, repetitions or duration, rest periods, weekly scheduling, warm-up and cool-down suggestions, and general safety guidance.

The main goal of FitBuddy is to make fitness planning simple and accessible. The system uses the user's stated goal and constraints to produce a more relevant plan. It is an educational/project prototype and is not a replacement for professional medical or fitness advice.

Keywords: Artificial Intelligence, Gemini, Generative AI, Fitness Planner, Personalized Workout, Prompt Engineering.

2. Introduction

Many people start exercising without knowing how to organize their weekly workouts. A fitness plan normally depends on the person's goal, experience, available time, equipment, and preferred activities. FitBuddy addresses this problem by using a generative AI model to convert user requirements into an easy-to-follow fitness plan.

3. Problem Statement

Beginners may find it difficult to select suitable exercises, decide workout frequency, determine repetitions and rest periods, and organize exercises into a weekly schedule. A fixed plan cannot easily adapt to different goals and constraints. FitBuddy provides a simple system that accepts user inputs and generates a personalized, structured fitness plan.

4. Objectives

• Create a simple user-friendly fitness planning application.

• Collect basic fitness goals and workout preferences.

• Use Gemini to generate personalized fitness recommendations.

• Produce a clear weekly workout schedule.

• Include warm-up, cool-down, rest, and general safety guidance.

• Reduce the time required to manually design a basic workout plan.

5. Proposed System

FitBuddy follows an input → AI processing → output workflow. The user enters details through a form. The application validates the required fields and creates a prompt. The prompt is sent to Gemini through the configured API/service layer. Gemini returns a structured plan, which the application formats and displays.

System Workflow

Step	Process
1	User enters fitness information and preferences.
2	Application validates and organizes the input.
3	Prompt is created with clear instructions for Gemini.
4	Gemini generates a personalized fitness plan.
5	Application formats the response into sections.
6	User views or saves the generated plan.

6. Main Modules

1. User Input Module: Collects age, goal, experience level, equipment, workout days, session duration, and preferences.

2. Validation Module: Checks required fields and reasonable input values.

3. Gemini AI Module: Sends a structured prompt to Gemini and receives the generated plan.

4. Plan Formatter: Converts the response into headings, daily schedules, exercises, and safety notes.

5. User Interface: Displays the generated plan clearly.

6. Save/Export Module: Optional module for saving the generated plan.

7. Input Parameters

Parameter	Example
Age	20
Fitness goal	General fitness
Experience	Beginner
Workout days	4 days/week
Session duration	45 minutes
Equipment	Dumbbells and exercise mat
Preference	Home workout

8. Gemini Prompt Design

Prompt engineering is important because the quality and structure of the generated plan depend on the instructions given to the model. FitBuddy asks Gemini to act as a fitness-planning assistant and return information in a predictable structure.

Example prompt:

```
You are FitBuddy, an AI fitness planning assistant. Create a beginner-friendly weekly
workout plan using the following user information:

Age: 20
Goal: General fitness
Experience: Beginner
Days available: 4 days/week
Session duration: 45 minutes
Equipment: Dumbbells and mat
Preference: Home workout

Return the plan with: warm-up, daily exercises, sets/repetitions or duration, rest periods,
cool-down, weekly rest days, and general safety guidance. Keep the plan clear and
practical. Do not diagnose medical conditions or provide medical treatment.
```

9. Sample Generated Plan

Day	Workout	Example
Monday	Full Body	Squats, dumbbell press, rows, lunges, plank
Tuesday	Rest / light activity	Walking and mobility
Wednesday	Upper Body	Push-ups, dumbbell rows, shoulder press, curls
Thursday	Rest	Recovery
Friday	Lower Body + Core	Squats, Romanian deadlift, lunges, plank
Saturday	Optional activity	Light walking/stretching
Sunday	Rest	Recovery

The exact exercises and volume should be generated according to the user's stated information and reviewed for suitability. The application should encourage users to stop if they experience pain or unusual symptoms and seek qualified professional advice when appropriate.

10. Technology Stack

Layer	Suggested Technology
Frontend	HTML, CSS, JavaScript or a simple Python UI
Backend	Python / Flask or FastAPI
AI Model	Google Gemini model through the appropriate Gemini API/SDK
Data	JSON / SQLite for a simple prototype
Output	HTML page or PDF export

11. Basic Implementation Logic

1. Create the fitness input form.
2. Read and validate user input.
3. Build a prompt containing the user information and output requirements.
4. Send the prompt to the configured Gemini API using a securely stored credential.
5. Receive the generated response.
6. Display the plan in a structured format.
7. Allow the user to regenerate or save the plan.

12. Example Pseudocode

```
START
Read user details
Validate input
Create Gemini prompt
Send prompt to Gemini
Receive AI response
Format workout plan
Display plan to user
END
```

13. Advantages

• Personalized output based on user-provided preferences.

• Fast generation of a structured weekly plan.

• Easy-to-use interface for beginners.

• Can be adapted for different goals and equipment.

• Can be extended with progress tracking and plan regeneration.

14. Limitations and Safety

Generative AI can produce incorrect, unsuitable, or overly general recommendations. FitBuddy should therefore be treated as an educational planning tool rather than a medical or clinical system. It should not diagnose injuries or diseases or provide treatment. Users with medical conditions, injuries, pregnancy-related concerns, or other special circumstances should consult an appropriate qualified professional.

The application should collect only information necessary for the feature, protect API credentials on the server side, and avoid exposing private user data in prompts, logs, or error messages.

15. Future Enhancements

• User login and workout history.

• Progress tracking with charts.

• Automatic plan adjustment based on completed workouts.

• Nutrition-planning features with appropriate safety limits.

• Exercise video demonstrations.

• Voice-based interaction.

• Wearable-device integration.

• PDF and calendar export.

16. Conclusion

FitBuddy demonstrates how generative AI can be used to create a simple personalized fitness-planning application. By combining a user-friendly input form, prompt engineering, and a Gemini model, the system can transform user requirements into a structured weekly workout plan. The project provides a practical example of AI in a real-world application while highlighting validation, privacy, and responsible use.

17. Viva / Presentation Questions

Question	Short Answer
What is FitBuddy?	An AI-based fitness plan generator.
Why is Gemini used?	To generate personalized natural-language fitness plans from user inputs.
What is prompt engineering?	Designing clear instructions and context to obtain useful model output.
What are the main inputs?	Goal, age, experience, schedule, duration, equipment, and preferences.
Is the plan medical advice?	No. It is a general AI-generated planning aid.